Question ID: 781c2f6e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

The function f is defined by $f(x) = a\left(2.2^x + 2.2^b\right)$, where a and b are integer constants and $0 < a < b$. The functions g and h are equivalent to function f, where k and m are constants. Which of the following equations displays the *y*-coordinate of the *y*-intercept of the graph of $y = f(x)$ in the *xy*-plane as a constant or coefficient?

I. $g(x) = a(2.2^x + k)$
II. $h(x) = a(2.2)^x + m$

Answer

A. I only

B. II only

C. I and II

D. Neither I nor II

Correct Answer: D

Rationale

Choice D is correct. A *y*-intercept of a graph in the *xy*-plane is a point where the graph intersects the *y*-axis, or a point where $x = 0$. Substituting 0 for x in the equation defining function f yields $f(0) = a\left(2.2^0 + 2.2^b\right)$, or $f(0) = a\left(1 + 2.2^b\right)$. So, the *y*-coordinate of the *y*-intercept of the graph is $a\left(1 + 2.2^b\right)$, or equivalently, $a + a(2.2)^b$. It's given that function g is equivalent to function f, where $0 < a < b$. It follows that $k = 2.2^b$. Since $a(2.2)^b$ can't be equal to 0, the coefficient a can't be equal to $a + a(2.2)^b$. Since $0 < a$, the constant k, which is equal to 2.2^b, can't be equal to $a + a(2.2)^b$. Therefore, function g doesn't display the *y*-coordinate of the *y*-intercept of the graph of $y = f(x)$ in the *xy*-plane as a constant or coefficient. It's also given that function h is equivalent to function f, where $0 < a < b$. The equation defining f can be rewritten as $f(x) = a(2.2)^x + a(2.2)^b$. It follows that $m = a(2.2)^b$. Since $a(2.2)^b$ can't be equal to 0, the coefficient a can't be equal to $a + a(2.2)^b$. Since $0 < a$, the constant m, which is equal to $a(2.2)^b$, can't be equal to $a + a(2.2)^b$. Therefore, function h doesn't display the *y*-coordinate of the *y*-intercept of the graph of $y = f(x)$ in the *xy*-plane as a constant or coefficient. Thus, neither function g nor function h displays the *y*-coordinate of the *y*-intercept of the graph of $y = f(x)$ in the *xy*-plane as a constant or coefficient.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.